# Homework 6. Jackknife, Boostrap

Stat 427/627, Statistical Machine Learning See

due date on Canvas.

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*16 pts.* This assignment covers Jackknife and Bootstraping.

## 1 Basses and sopranos (Jackknife) (6 pts)

An acoustic studio needs to estimate the range of voice fundamental frequencies that an adult singer can produce. A sample of $n = 10$ recordings contains frequencies 102, 115, 127, 127, 162, 180, 184, 205, 239, 240.

(a) Manually compute (by hand) the jackknife estimator of the population lowest fundamental frequency of a human voice. Compare your results with the natural range of human voice frequencies. (Use Google or Wiki.)

(b) Use software to confirm your result.

(c) (Stat-627 only) Generalize the results. Assume a sample $X_1, \ldots, X_n$ of size $n$, where $X_1, X_2$ are the smallest two observations. Also assume there is no tie. Derive the equation for the Jackknife estimator of the population minimum. Use your result in (a) to verify your formula.

## 2 Ex.5.4.9. Bootstrap the mean of median house values in the `Boston` dataset. (p. 223, 10 pts)

We will now consider the `Boston` housing data set from the {`MASS`} library.

(a) In this data set, variable `medv` is the median value of owner-occupied homes in \$1,000s. Based on the `Boston` data set, provide an estimate for the population mean $\mu$ of `medv`. Call this estimate $\hat{\mu}$.

(b) Estimate the standard error of $\hat{\mu}$ (as we know, $\text{StdError}(\bar{x}) = s/\sqrt{n}$, where $s$ is the sample standard deviation. R function `sd()`.)

(c) Estimate the standard error of $\hat{\mu}$ using the bootstrap method. How does this compare to your answer from (b)?. Remember to set your seed to be reproducible.

(d) Based on your bootstrap estimate from (c), provide a 95% confidence interval for $\mu$. A popular approximation is $\hat{\mu} \pm 2\text{StdError}(\hat{\mu})$. Compare it to the results obtained using an R command `t.test(Boston$medv)`.

(e) Now, estimate $M$, the population median of `medv` with the sample median $\hat{M}$.

(f) We now would like to estimate the standard error of $\hat{M}$, but unfortunately, there is no simple formula for computing the standard error of a sample median. Instead, estimate this standard error using the bootstrap method.

—— **This is the end of HW 6.** ——

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